Rose Martin
Professor Dasari
Fundamentals of Networks
10 April 2026

Programming Assignment 2026

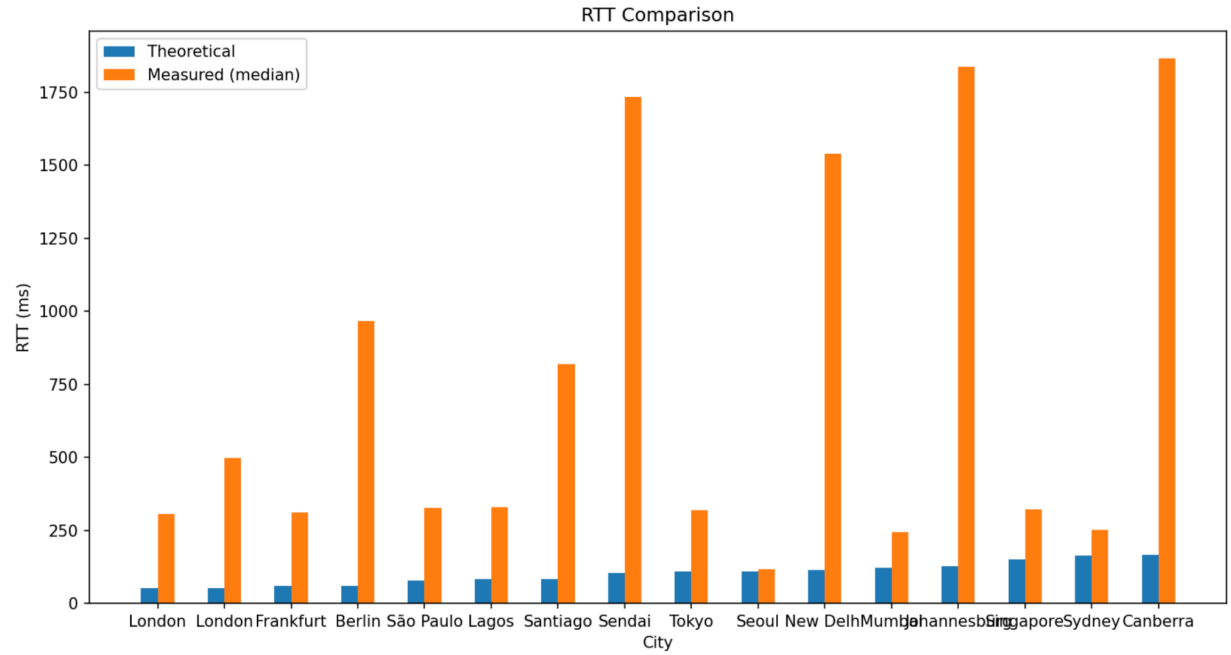
Goal: Measure real network latency to cities around the world and compare it against the physical limits of fiber optics.

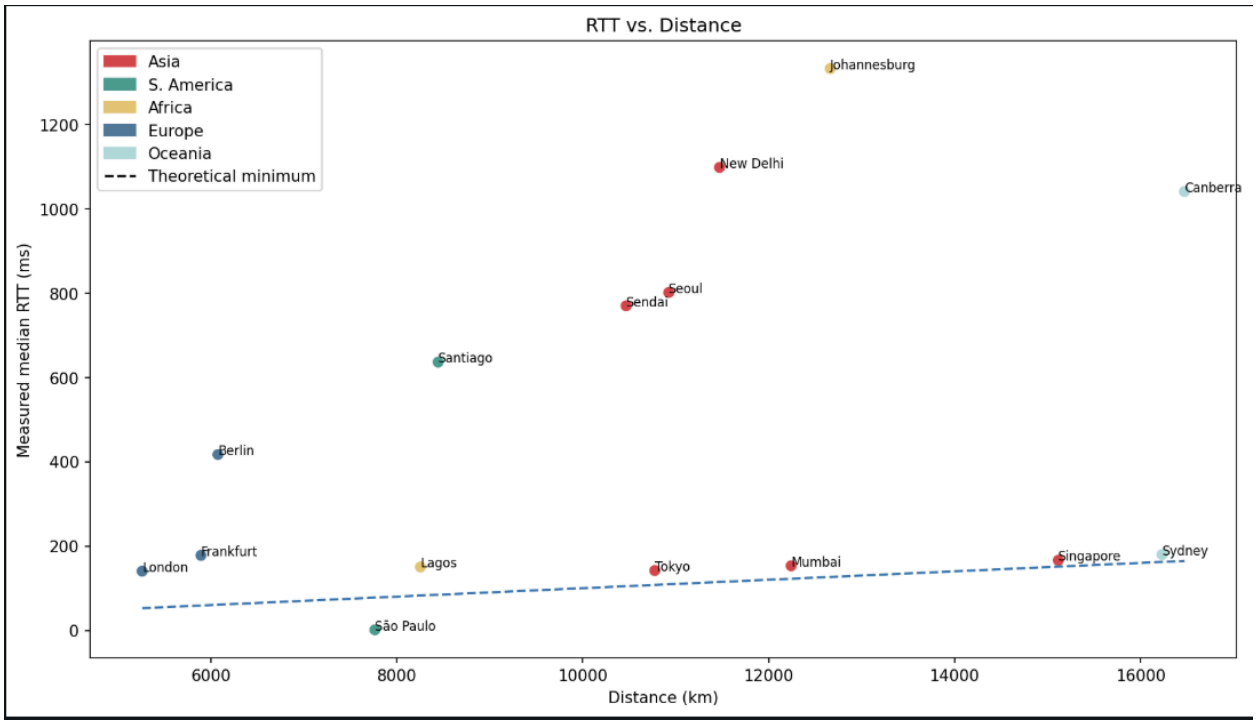
Terminal Table Screenshot:

```
Your location: Burlington (42.5048, -71.1956)
```

City	Dist km	Median ms	Theor. ms	Ratio
London	5263	138.3	52.6	2.63
Frankfurt	5896	189.8	59.0	3.22 ⚠️
Berlin	6078	418.3	60.8	6.88 ⚠️
São Paulo	7765	1.2	77.7	0.01
Lagos	8255	120.9	82.6	1.46
Santiago	8446	637.3	84.5	7.55 ⚠️
Sendai	10467	766.5	104.7	7.32 ⚠️
Tokyo	10775	140.0	107.7	1.30
Seoul	10925	802.1	109.2	7.34 ⚠️
New Delhi	11471	1124.0	114.7	9.80 ⚠️
Mumbai	12241	135.7	122.4	1.11
Johannesburg	12660	1274.6	126.6	10.07 ⚠️
Singapore	15113	181.9	151.1	1.20
Sydney	16229	185.9	162.3	1.15
Canberra	16469	1039.6	164.7	6.31 ⚠️

Figures 1 and 2:





Analysis Questions:

Answer three questions:

1. Which city has the highest inefficiency ratio? Look it up on submarinecablemap.com — what cables serve it and how does that explain your result?

The city with the highest inefficiency ratio was Johannesburg. Johannesburg is not a location that serves as an international cable hub. Therefore, traffic and delays are increased because Johannesburg is located inland and must be connected from Africa through Europe. This indirection can cause congestion and queuing delays.

2. Which city is closest to the theoretical minimum? What does that tell you about routing infrastructure there?

The city that is closest to the theoretical minimum is Sao Paulo. The routing infrastructure there is direct and efficient. Data can take a direct path with limited network congestion/delay.

3. Your packet to Lagos almost certainly routes through Europe first. Why does Africa route through Europe, and what would need to change to fix it?

Lagos routes through Europe first. Africa routes through Europe first because there are limited trans-Atlantic submarine lines from Africa. No lines run from North America to Africa. Fixing this problem would involve laying more direct submarine cables to connect Africa to the other continents.